

# SENTIMENT CLASSIFICATION OF E-COMMERCE PRODUCT REVIEWS USING LONG SHORT-TERM MEMORY

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## ARTICLE INFO

*Keywords:*

LSTM

e-commerce reviews

*natural language processing*

sentiment classification

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*Article history:*

*Received: Dec 31<sup>st</sup>, 9999*

Revised: Dec 31<sup>st</sup>, 9999

Accepted: Dec 31<sup>st</sup>, 9999

Available online: Dec 31<sup>st</sup>, 9999

## ABSTRACT

Customer reviews on e-commerce platforms are an important source of information for sellers to understand consumer satisfaction with the products sold. However, the growing number of reviews posted every day makes manual analysis inefficient. This study proposes the use of Long Short-Term Memory (LSTM) to classify sentiment in e-commerce product reviews into three classes: positive, neutral, and negative. The dataset used consists of 10,000 reviews collected from a popular e-commerce platform in Indonesia. The research stages include text preprocessing, feature extraction using word embedding, and training the LSTM model with several hyperparameter variations. Testing results show that the proposed LSTM model achieves an accuracy of 89.4%, higher than the comparison method, Support Vector Machine (SVM), which only achieves 82.1% accuracy. These results indicate that LSTM is effective at capturing context and word relationships in informal Indonesian-language reviews. This research is expected to help business owners automatically and continuously monitor product reputation in real time.

## 1. Introduction

The rapid growth of online transactions in Indonesia has led to an increasing volume of customer reviews on various e-commerce platforms [1]. These reviews contain consumer opinions that sellers can leverage to evaluate product quality and service. Sentiment analysis has become one of the automatic solutions for processing large volumes of opinion data without requiring manual reading of each entry [2]. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequaleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

## 2. Literature review

Previous studies related to sentiment analysis are summarized in. Machine-learning-based classification methods such as Naive Bayes and SVM have been widely used, but they have limitations in capturing the context of long sentences. A comparison of model architectures is shown in Table 1. The general equation of the sigmoid activation function used in the LSTM gates is shown in (1).

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (1)$$

### 2.1. LSTM Method

LSTM is an extension of the Recurrent Neural Network (RNN) designed to address the vanishing gradient problem through a gating mechanism, namely the forget gate, input gate, and output gate. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

### 3. Methodology

This study uses a review dataset that has been manually labeled with sentiment by three annotators. The preprocessing stages include case folding, tokenization, stopword removal, and stemming using the Sastrawi library. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

### 4. Results and Discussion

The testing results show that the best model performance is achieved at epoch 25 with a learning rate of 0.001. A comparison of accuracy across models is presented in the following table. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

### 5. Conclusion

Based on the testing results, the LSTM method proves superior to SVM in classifying sentiment in Indonesian-language e-commerce reviews, achieving an accuracy improvement of 7.3%. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaeque doleamus animo, cum corpore dolemus, fieri.

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